

Institute of Actuaries of India

Subject SA5 – Finance

October 2014 Examination

INDICATIVE SOLUTION

Solution 1:**(i) Capital Adequacy**

- For any financial institution, capital plays an important role acting as a buffer against potential future losses. It thereby protects the customers of the financial institution against the risk of the insurer going insolvent, in which case they would lose some or all of their savings.
- Capital adequacy refers to the amount of capital required to cover the potential diminution in the value of assets and other exposures over a given period of time at a given level of confidence.
- Capital adequacy for a bank is typically expressed as a ratio of its capital to risk weighted assets
- Banks are expected to keep adequate capital for the following reasons :
 - Banks need to keep adequate capital to underpin the risk of loss in the value of exposures, businesses, etc so as to protect depositors and general creditors against loss
 - Adequate capital is necessary to maintain the confidence of the depositors in the healthy state of the bank. If the depositors have doubts in this regard, then there is likely to be a run on deposits which will undermine the financial position of the bank.
 - If the bank has insufficient capital relative to the risks faced, then the bank will have to either [a] raise more capital at short notice with the consequent dilution of returns available to the shareholders; or [b] rein in its lending and investment activities with a consequent reduction in profitability.
 - Maintaining adequate capital places the bank in a better position to withstand any future problems in the international financial markets.
 - In particular banks need to have adequate capital to act as a buffer against their lending activities. Banks are more vulnerable to such losses in times of general economic difficulty when their borrowers are more likely to default.

(ii) Key Indicators

- Domestic Indicators of Economic and Price Levels [1 mark]
 - Real & Nominal GDP
 - Unemployment Rate
 - Nominal Disposable Personal Income
 - Consumer Price Index and/or Wholesale Price Index
- Domestic Indicators of Asset Prices and Financial Market Conditions
 - National House Price Index
 - National Commercial Real Estate Price Index
 - BSE/NSE Stock Market Index
 - BSE/NSE Stock Market Volatility Index
- Domestic Indicators of Interest Rates
 - Treasury Bill Rates
 - Yield on long dated government bonds [e.g., 10 year GSec]
 - Yield Spread on AAA rated and BBB rated Corporate Bonds of the same tenor as the long dated government bonds
 - Interest rate on conventional fixed rate long term [e.g., 30 year] mortgages
- International Macro- Economic Indicators
 - Growth rate in real GDP
 - Percentage Change in the Consumer/ Retail Price Index or any other local equivalent index
 - INR / Foreign Currency Exchange Rate

The above international macro- economic indicators need to be considered for each country which is regarded as a significant trading partner or as a country which can significantly impact the Indian economy.

(iii) Components of the Capital Plan of a Bank

- Break-Up of the current capital into
 - Tier 1 Capital which includes the core capital items like shareholders' equity and disclosed reserves
 - Tier 2 which includes supplementary capital items like revaluation reserves, general provisions, hidden reserves, long term subordinated debt and other approved capital items
 - Tier 3 capital which includes capital items such as short term subordinated debt.
- Assessment of the expected sources and expected uses of capital over a planning horizon of, say, 2 to 3 years- taking into account the bank's complexity, size, risk profile and scope of operations- under both expected and stressful conditions
- Detailed description of the bank's process for assessing capital adequacy, including a discussion of how under expected and stressful conditions it will [i] maintain sufficient capital commensurate with its risks; [ii] maintain regulatory capital above the minimum prescribed capital ratios; and [iii] serve as a source of strength to its subsidiary depository institutions
- Assessment of its financial flexibility particularly in terms of its ability to readily access sources of capital under both expected and stressful conditions
- Assessment of the expected changes to the bank's business and risk profile
- Description of the bank's current and future dividend policy

[5]**(iv) Calculating VaR Using Historical Simulation Approach**

- The first step will be to describe each trading asset in terms of the chosen "market factor". For example the market factor can be 3-month interest rates.
- The next step will be to collect historical data on the chosen market factor over a selected period in the past. For example, we can collect the data on the changes in the 3-month interest rate for 99 day periods over, say, the last 10 years. These 99 day periods can be over non- overlapping or overlapping intervals.
- We can then calculate the changes in the market value of the relevant trading asset when subjected to the changes in the market factor experienced in the past [using the data collected in the previous step]

- We could fit a distribution to the results obtained in the previous step [e.g. normal distribution]. Then we can calculate the mean and standard deviation of this distribution.
- As the next step, we calculate the start of the one-sided, say, 0.5% tail of the distribution. For example, if we had chosen the normal distribution, we will calculate the mean less 2.576 standard deviations.
- The difference between the value of the asset calculated at the start of the tail [as explained in the previous step] and the current market value of the asset will be the required VaR number.

[6]

(v) Supervisory Stress testing & Capital Adequacy Review: Key Principles

- The main purpose of introducing a supervisory stress testing framework is to provide a quantitative forward looking assessment of the capital adequacy of banks.
- Regular supervisory stress tests aim to ensure that the banks' are adequately capitalised to continue operations throughout times of economic and financial stress.
- Supervisory stress testing is intended to bolster public confidence in the long-run stability of the banking system by demonstrating the range of severe but plausible stresses that the regulator [RBI] expects banks to withstand.
- The supervisory stress testing framework aims at strengthening a bank's own capabilities to identify and quantify the relevant risks. This is done by evaluating the bank's portfolio under two distinct set of scenarios: common scenarios applicable to all banks and bespoke scenarios which unique to the bank. This would enable the bank to identify and measure the risks which can have the maximum negative impact.
- The stress testing framework must ensure that the results of the stress tests result in credible policy actions both at a system-wide level [at the level of the banking sector] and at the level of an individual bank.
- The results of the stress tests will form an objective basis for evaluating and approving/ rejecting the capital distribution plans proposed by the bank.

[5]

(vi) Buffer Concepts under Basel III:

- Counter Cyclical Buffer: This buffer will increase the capital requirement at the peak of a credit cycle and reduce the capital requirement during the downtrend of the credit cycle.

This buffer will range from 0% to 2.5% of the risk weighted assets and will consist of fully loss-absorbing capital items like equity.

- Capital Conservation Buffer: This buffer will ensure that the bank maintains a capital cushion to absorb losses during periods of financial and economic stress.

This buffer is likely to be 2.5% of the risk weighted assets.

- Buffer for Systemically Important Financial Institutions: This buffer will ensure that large banks pose a lesser threat to the financial system. Such banks are expected to have loss-absorbing capability beyond the Basel III requirements.

This buffer will be in the forms of capital surcharges, contingent capital and bail-in debt.

[3]

(vii) Key Features of Basel III

- Basel III prescribes higher capital targets for banks and aims at improving the quality of capital held by the banks. This is done by adding a capital conservation buffer and by increasing the overall Tier 1 capital requirement from the current minimum of 4% to 6% of the risk weighted assets.
- A review of the financial crisis of 2008 revealed that the value of assets tend to fall steeper than what would be expected based on historical experience. Hence Basel II proposes a leverage ratio that limits the amount of loans a bank can have regardless of how risky or otherwise the loans are.
- Basel III introduces a non-risk capital based liquidity coverage ratio to ensure that the banks are well positioned to meet short term obligations under conditions of stress.
- Basel III adds a counter cyclical buffer to increase the capital requirement at the peak of the credit cycle and a draw down from this buffer during the downtrend of the credit cycle.

- Basel III adds an additional capital requirement for systemically important financial institutions to ensure that the large banks pose less of a threat to the overall stability of the banking system.
- Basel III requires the banks to maintain a minimum Net Stable Funding Ratio [NSFR] in order to ensure that the banks increase their reliance on stable sources of long term funds for financing the assets.

[5]

(viii): Measuring and Managing Liquidity

- An important measure for assessing liquidity risk is the “net liquid assets” or the “liquidity gap” which is measured as the difference between the liquid assets and volatile liabilities.

This is usually expressed as a ratio of Net Liquid Assets to Volatile Assets. A ratio of less than 1 indicates the possibility of a liquidity problem.

While analysing the liquid assets position, allowance needs to be made for the liquidation costs associated with converting the “near liquid items” into cash. These costs will be a function of the brokerage expenses, bid-offer spread as well as the time available for conversion.

A six month maturity criterion is typically adopted for classifying assets and liabilities as “liquid assets” and “volatile liabilities”.

- Another important measure for assessing liquidity risk is the “liquidity risk elasticity [LRE] which is measured as follows:

[i] Calculate the present values of assets [A] and liabilities [L] using the cost of funds as the discount rate. Therefore market value of equity $E = A - L$

[ii] Measure the change in the market value of equity arising from a change in the cost of funds. This measure is called the LRE.

If $LRE = 0$, then the bank has zero liquidity risk. If the LRE is sharply negative the bank will to shorten the maturity of the assets and/or lengthen the maturity of the liabilities in order to mitigate the liquidity risk.

- The other commonly used measures for assessing liquidity risk [as proposed by RBI] are as follows:
 - Core Deposits to Total Assets Ratio: This ratio measures the extent to which assets are funded through a stable deposit base.

- [Loans +Mandatory SLR + Mandatory CRR +Fixed Assets] to Total Assets Ratio: If this ratio is high, then it indicates “illiquidity” in the balance sheet of the bank. This is because fixed assets, loans, mandatory cash reserves and statutory liquid investments are the least liquid assets of a bank

Note: RBI defines Core Deposits, liquid assets and volatile liabilities as follows:

- ✓ Core Deposits = All deposits with a remaining maturity of more than 1 year
- ✓ Volatile Liabilities = Deposits , borrowings and bills payable with a remaining maturity of less than 1 year
- ✓ Liquid Assets = Cash +Excess CRR balances with RBI+ Balances with banks + Bills purchased and discounted with a remaining maturity up to 1 year + Investments with a remaining maturity up to 1 year

[5]

(ix) a: Methods for Converting Off- balance sheet items

- The two methods for covering off-balance sheet items like derivative exposures into on balance credit equivalents are:
 - Original exposure method
 - Current exposure method
- **Original Exposure Method :**
 - Under this method, the notional amount of the derivative contracts is multiplied by a chosen % according to the original term of the derivative contract and the type of the underlying asset [e.g., interest rate, exchange rate, etc].

This approach will result in an amount which represents the maximum actual value of the derivative contract during its life.
 - The main advantage of the original exposure method is that it is a quick and easy method of assessing exposure to credit risk.
 - The disadvantage of the original exposure method is that it is a broad brush method which results in a very approximate assessment of the risk.
- **Current Exposure Method :**
 - This method takes the current market value of the derivative contract and makes an adjustment in the form of an “add on”. The “add-on” reflects the potential maximum increase in value till the maturity date of the contract.

The “add-on” is expressed as a% of the notional amount and is based on the remaining term to maturity & not on original term. This approach will result in an amount which represents the maximum actual value of the derivative contract during its life.

- Under this approach only positive market values are included and negative values are ignored. This is because negative market value of a derivative contract means that it is a liability for the bank and therefore represents no credit risk to the bank. However the “add-on” part of the computation is applied to all derivative contracts.
- The main advantage of the current exposure method is that it is more realistic and it will give a lower capital requirement than what is calculated under the original exposure method.

[7]

(b): Control Mechanisms related to Derivative Positions

- The Board of Directors of the bank must review all management decisions related to the appropriateness and effectiveness of derivative strategies and positions.
- The Board must also ensure that the employees involved in derivative trading and related operations have the necessary skills and experience.
- The risk analysis processes for derivative activities must include identifying the risk, estimating its significance and assessing the likelihood of its occurrence.
- The Board and the management must obtain sufficient and timely information for the purpose of monitoring the strategies involving the use of derivative instruments.
- The internal control framework for monitoring derivative activities must include [i] clear segregation of duties like segregating trading activities from account settlement; [ii] controls related to managerial oversight and responsibilities ; [iii] operational controls like dealing limits ; and [iv] having well defined reporting processes.
- The control systems relating to derivative activities must be monitored to ensure the integrity of system-generated reports.

[6]

(x): Operational and Accounting Information

A. Accounting Information :

- Values of assets and liabilities on a fair-value basis
- Accounting treatment for operating and financial leases

- Accounting treatment for securitised assets particularly for those securitised assets which present a residual exposure to the bank
- Accounting treatment for consolidating subsidiaries in the financial statements of the holding company
- Legal liability on the part of the bank to pay claims of its subsidiaries
- Tax liabilities outstanding
- Funding position of the employee benefit plans operated by the bank like the pension plan and gratuity plan
- Off balance sheet items like stand-by letters of credit, outstanding financial derivative contracts such as forwards, futures, options and swaps, etc. Each of these items needs to be described in terms of the original term, outstanding term, and market value and credit conversion factors.
- Investments split by asset category in order to assess the likely market risk of the investment portfolio.
- Risk evaluation models used and the risk capital held against risks like default risk, market risk and operational risk

B. Operating Information :

- Lending and deposit portfolios of the bank classified under the heads of retail and wholesale banking with examples
- Track record of the Board and the Senior Leadership Team
- Risk management policies and procedures
- Audit coverage

[10]

[63 Marks]

Solution 2:

(i)

- The current inflation rate environment in India is characterised by a downtrend in the inflation (CPI inflation fell from a peak of 11% pa in November 2013 to about 7.8% by August 2014) caused by a fall in food price inflation, fuel price inflation and housing price inflation. It is expected that this downtrend will continue and overall inflation rate may be around 6% in 2016.

- There has been an easing of monetary conditions. The money market rates as well as long term yields on government bonds and high quality corporate bonds have declined during the last 4 to 5 months (since August 2014). If RBI reduces the repo rate and inflationary expectations remain consistent with the current downtrend in inflation, then the government bond yields are likely to decline further.
- The trends in the global interest rates- particularly the interest rate environment in the US – will have a bearing on an interest rate trends in the Indian market. If US interest rates increase and Rupee depreciates against USD, RBI may be compelled to keep the policy rates unchanged if not increase these rates.
- The current account deficit and the ability of the government to meet the fiscal targets will have a bearing on the level of the long term interest rates.

[4]

(ii)

- | | |
|---|--------|
| i. Face Value of the bonds to be called | (600) |
| ii. Call Premium (0.04 * 600) | (24) |
| iii. Face Value of the bonds to be issued | 600 |
| iv. Issue costs of the proposed bond issue | (6) |
| v. Tax shield on the call premium and issue cost (30 * 0.35) = 10.5 | |
| vi. Net Initial outlay = i+ii+iii+iv+v | (19.5) |
| vii. Savings in interest cost per annum (0.02 * 600) =12 | |
| viii. Tax on savings in interest (0.35*12) = (4.2) | |
| ix. Post tax savings in interest cost per annum = 7.8 | |
| x. Present value of post-tax savings in interest cost | |
| 7.8*a (6) @7.5% =7.8*4.694 =36.6 | |
| xi. Net present value =36.6-19.5 = 17.1 | |

Conclusion: Since the NPV of the decision is greater than zero, Cadmin can go ahead with the decision to call back the 12% bonds.

[4]

(iii) Cash flows on a Callable Bond Vs Cash flows on an ABS:

- A callable bond is a single security while the ABS consists of many underlying assets, each with a prepayment option
- Even if the interest rates were to fall immediately after the issue of the callable bond, the issuer can exercise the call option only on a fixed (predetermined) future date. On the other hand, the prepayments on an ABS will increase immediately after the fall in interest rates.
- The callable bond will have a zero rate of prepayment when the market interest rate is greater than the coupon rate. In the case of an ABS, a part of the principal amount is paid along with each interest payment. This makes the duration of the ABS less than the duration of the callable bond.
- Default by the issuer of a callable bond will impact the callable bond either in full or not at all. In the case of ABS, the default experience of the underlying assets will have the impact of gradually reducing the interest payments on the ABS.

[3]**(iv) Value of the Option Embedded in a Callable Bond**

- To calculate the value of this option, we need to have a Monte Carlo or Binomial Model to simulate many possible future interest rate paths. This is necessary because the future cash flows are not known with certainty.
- The future interest rate paths are generated using an arbitrage free stochastic model of the future term structure of interest rates
- Each path will predict the long term and short term interest rates at ,say, annual intervals between the present date and the specified maturity date
- At each annual interval, the model will check whether, given the predicted interest rate at the time, the callable bond will be redeemed. This will happen if the predicted rate (refinancing rate) is less than the coupon rate on the callable bond. If the bond is not redeemed at that annual review date, then the model will check for redemption at the next annual review date and so on
- The present value of each path is therefore the discounted value of all the future cash flows on the path – coupons plus anticipated redemption payments.
- The value of the callable bond will be the average of present values of all the interest rate paths

- As the next step we need to calculate the value of the equivalent straight bond without an embedded call option.
- The value of the embedded call option = Value of the equivalent straight bond minus the Value of the callable bond

[6]

(v) NPV [IP-1] = $40v + 100v^2 + 100v^3 + 100v^4 - 300$ at 14%

$$= 35.1 + 76.9 + 67.5 + 59.2 - 300$$

$$= (-) 61.3$$

[1]

(vi) Value of the Option to Invest in IP-2

- We will assume that the follow on investment is made at the end of year 4
- We can use the Black – Scholes option pricing formula
(0.5 mark)

Step 1: Calculate the PV of the expected future cash flows from IP-2 at time 0

$$= (80v + 200v^2 + 200v^3 + 200v^4) \cdot v^4 = 477.4 \cdot 0.592 = 282.6$$

(0.5 mark)

Step 2: Calculate d_1 and d_2

$$d_1 = [\ln (282.6/600) + (0.09 \cdot 4) + (0.3 \cdot 0.3 \cdot 2)] / 0.6$$

$$= (-) 0.355$$

$$d_2 = d_1 - \sigma \cdot t^{0.5}$$

$$= -0.355 - 0.6$$

$$= (-) 0.955$$

Step 3: Calculate $N(d_1)$ and $N(d_2)$

$$N(-0.355) = 0.3613$$

$$N(-0.955) = 0.1698$$

Step 4: Plugging in the values in the B-S formula, we get

$$C = 282.6 \cdot 0.3613 - 600 \cdot e^{-0.36} \cdot 0.1698$$

$$= 102.10 - 71.08$$

$$= 31.0$$

Strategic NPV of the project IP

$$= - 61.3 + 31.0$$

$$= (-) 30.3$$

The strategic NPV is also negative which implies that the IP – 1 is not a worthwhile project even after taking into account the value of the real option

[4]

(vii) (a) Financial Options and Real Options

- The information required for valuing options and making decisions about exercising them is more readily available for financial options than for real options.
- While the right to exercise a financial option is unambiguous, the holder of the real option often is unclear about what the precise right is and how long the same will last

[2]

(b) Key Issues Connected with Valuation of Real Options:

- Incorrect Application of Option Pricing Models:
 - Option Pricing Models tend to assume constant variance. While such assumptions may apply to short term options on traded assets, they may not apply to long term options on non- traded assets
- Assumptions of a fixed exercise price: While the exercise price of a financial option is fixed, the exercise price of a real option tends to vary over time. (Example: cost of developing an oil field). Hence the assumption of a fixed exercise price in the valuation of real option is often incorrect.
- There is a tendency to overestimate the value of flexibility in the case of real (flexibility) options. This can happen when the option is not actually exercised on those exercise dates assumed in the option valuation model.
- Real options like investment timing options and flexibility options have value only if the firm has a certain degree of exclusivity. If the competitors can imitate the firm, real options may not be valuable.
- There can be a tendency on the part of project sponsor to invoke real options to justify a financially unviable project on strategic grounds

[4]

(viii)

- The main factor which would have been considered is the financial viability of the IP-1 project. The NPV of the IP-1 project is negative and the strategic NPV (taking into account the value of the option to make an additional investment) is also negative
- The sustainability and adequacy of the Cadmin's cash flows could be another factor influencing the rating of BBB
 - The rating agency is probably foreseeing a weakening debt servicing ability given the additional debt burden and inadequate cash flows from the IP-1 project

The other factors considered by the rating agency will include:

- The technology capabilities and investment in R&D for manufacturing speciality industrial paints
- The flexibility in manufacturing in terms of the ability to shift between different types of paints with ease
- Cost competitiveness
- Access to raw materials
- Volatility in the global and domestic prices of paints

[5]

(ix) The plausible factors considered by the CIO will include:

- IRDA Investment Regulation & Investment Policies
- Concentration Risk
- Asset Liability Matching
- Viability of the Project for which funding is sought (0.5 mark)
- Credit Risk of the Issuer
- Alternative Investment Opportunities

[4]

[37 Marks]
